Owen
Manley
CSC362

Machine Learning Assignment #1

1a.) Model 1 will generalize instances that aren't present within the dataset already because it is already consistent with the data provided.

1b.) The second model is overfitting the data because it is overcomplicating the Model. The cases that are 'default' in Model 2 are already covered in Model 1; so, it is just more work.

2a.) Because there are 4 different features in the model, there is 2^4 possible combinations. Which, end up being 16 possible model combinations.

2b.) To determine the amount of models that would be consistent we look at the risk factors (which are 4 in total), then the stroke risk which is 3. Doing some calculations ($3^4 = 81$) it can be determined that there are 81 possible combinations in total.

3.) I think 80% of the work is spent on Business Understanding, Data Understanding, and Data preparation because it is essentially trying to predict outcomes based on data. Because of that, it's a lot of data to analyze and there are lots of things to be cautious about when predicting data. Things like bias or underfitting/overfitting are things to steer away from. Then, the data just needs to be implemented into a model.